

ROBUSTNESS PROPERTIES OF THE MDF ANALYSIS  
OF MOMENT STRUCTURES

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**Summary:** In this paper we study robustness properties of the minimum discrepancy function (MDF) analysis of moment structures. Necessary and sufficient conditions for asymptotic chi-squaredness of the MDF test statistics and asymptotic efficiency of the corresponding estimators are given. Interrelationship with the standard least squares theory of linear models is discussed.

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## 1. INTRODUCTION

In recent years considerable attention has been attracted to so-called covariance or, more generally, moment structural models (see, e.g., survey papers by Jöreskog, 1981; Browne, 1982; Bentler, 1983, and references contained therein). Under the assumption of multivariate normality statistical inference of such models has been developed for the methods of maximum likelihood and generalized least squares (Jöreskog, 1970; Jöreskog and Goldberger, 1972; Anderson, 1973; Browne, 1974, 1982; Lee and

Bentler, 1980; Shapiro, 1983). Both of these methods can be considered as particular cases of the minimum discrepancy function (MDF) approach which we adopt here (cf. Browne, 1982, section 1.5). It is known that for an appropriately chosen discrepancy function the corresponding MDF test statistic is asymptotically chi-squared and MDF estimators are asymptotically efficient. In the case of multivariate normality such discrepancy functions are well known and routinely employed in computer programs. It also has been found that for a certain class of covariance structural models the usual normal theory MDF analysis retains the above-mentioned properties of asymptotic chi-squaredness and asymptotic efficiency when applied to data having a distribution from the elliptical family (Muirhead and Waternaux, 1980; Browne, 1982, 1984; Tyler, 1982, 1983). In general the properties of asymptotic chi-squaredness and asymptotic efficiency hold for correctly specified discrepancy functions (cf. Browne, 1984, p. 67). This fact is utilized in distribution-free methods for analysis of covariance structures (see Browne, 1982, 1984). Unfortunately correct specification of the discrepancy function often involves estimation of huge covariance matrices which even if available makes the required computational procedures unfeasible. It is therefore important to investigate conditions under which the standard normal theory procedures retain their useful asymptotic properties when applied in situations for which they were not originally intended.

The main purpose of this paper is to investigate conditions under which the standard or slightly modified MDF procedures result in a correct statistical inference for misspecified discrepancy functions. We extend and generalize some results presented in Shapiro (1986) to include cases of singular covariance matrices and inexact models. Robustness properties of

the MDF test statistics, difference test statistics and estimators will be discussed.

The theory presented is naturally interrelated with standard linear regression analysis. Conditions for optimality and validity of least squares theory of linear models have been discussed extensively in statistical literature. The interested reader is referred to survey papers of Zyskind (1967), Watson (1967) and in particular to Rao and Mitra (1971, chapter 8). We shall utilize some relevant results of this theory while studying asymptotic efficiency of MDF estimators.

## 2. PRELIMINARY DISCUSSION

In this section we briefly outline statistical framework for the MDF analysis. Let  $\xi = (\xi_1, \dots, \xi_m)'$ , be a vector variable representing a parameter vector of some statistical population. For example in the analysis of covariance structures  $\xi$  will represent the elements of a  $p \times p$  covariance matrix  $\Sigma$ . That is  $\xi = \text{vec}(\Sigma)$ , where  $\text{vec}(\Sigma)$  denotes the  $p^2 \times 1$  vector formed by stacking columns of  $\Sigma$ . A structural model for  $\xi$  is an  $m \times 1$  vector-valued function  $g(\theta)$  which relates the parameter vector  $\theta$ ,  $\theta \in \mathbb{R}^q$ , to  $\xi$ :

$$(2.1) \quad \xi = g(\theta).$$

In some cases the model (2.1) is subject to equality constraints  $c_i(\theta) = 0$ ,  $i=1, \dots, k$ , on the parameters. We shall refer to such a model as a restricted model to distinguish it from the unrestricted model (2.1). It will be assumed that functions  $g(\theta)$  and  $c(\theta) = (c_1(\theta), \dots, c_k(\theta))$  are continuously differentiable.

Let  $\xi_0$  be the population (true) value of  $\xi$ . The model

(i.e. the null hypothesis) is said to hold if there exists a vector  $\theta_0 \in \mathbb{R}^q$ , in the restricted case satisfying  $c_i(\theta_0) = 0$ ,  $i=1, \dots, k$ , such that  $\xi_0 = g(\theta_0)$ . In order to investigate asymptotic power of test statistics we consider inexact models (alternative hypothesis). Consider a sample of size  $n$  and let  $\xi_{0,n}$  be the corresponding population value of  $\xi$  depending on the sample. It will be assumed that as the sample size  $n$  tends to infinity,  $\xi_{0,n}$  converges to a point  $\xi_0$  satisfying the model,  $\xi_0 = g(\theta_0)$ , in such a way that  $n^{1/2}(\xi_{0,n} - \xi_0)$  tends to a vector  $\mu$ . This is a standard assumption of population drift (cf. Stroud, 1972; Steiger *et al.*, 1985).

Let  $\hat{x}$  be a sample estimate of the population value  $\xi_{0,n}$ . For example in the analysis of covariance structures  $\hat{x}$  is usually given by  $\text{vec}(S)$  with  $S$  being the sample covariance matrix. We assume that  $n^{1/2}(\hat{x} - \xi_{0,n})$  has asymptotically a multivariate normal distribution with a null mean vector and a covariance matrix  $\Gamma$ . This implies that asymptotically  $n^{1/2}(\hat{x} - \xi_0)$  is  $N(\mu, \Gamma)$ . Given  $\hat{x}$  one fits the model (2.1) by minimizing the discrepancy between  $\hat{x}$  and  $\xi = g(\theta)$ , which is measured by means of a discrepancy function  $F(x, \xi)$  (cf. Browne, 1982). Then the (unrestricted) MDF test statistic is given by  $\hat{nF}$ , where  $\hat{F}$  is the (unrestricted) minimum of the discrepancy function

$$\hat{F} = \min_{\theta} F(\hat{x}, g(\theta)).$$

The associated MDF estimator  $\hat{\theta}$  is chosen to be a corresponding minimizer of  $F(\hat{x}, g(\cdot))$  over  $\mathbb{R}^q$ . Similarly in the case of a

restricted model the MDF test statistic is  $n\hat{F}$ , where  $\tilde{F}$  is the minimum of  $F(\hat{x}, g(\theta))$  subject to constraints  $c_i(\theta) = 0$ ,  $i=1, \dots, k$ .

Following Browne (1984) we say that the discrepancy function is *correctly specified* if  $V_0 = r^{-1}$ . Here  $V_0$  is the weight matrix associated with the discrepancy function  $F$  which is given by the matrix  $\frac{1}{2}(\partial^2/\partial\xi\partial\xi')F(\xi_0, \xi_0)$  of second-order partial derivatives (Shapiro, 1985a). For correctly specified discrepancy functions the following useful properties of the test statistics hold under mild regularity conditions:

- (i) The MDF test statistics  $n\hat{F}$  and  $n\tilde{F}$  are asymptotically chi-squared with  $m-q$  and  $m-q+k$  degrees of freedom and noncentrality parameters  $\delta$  and  $\delta^*$ , respectively.
- (ii) The difference test statistic  $n(\hat{F}-\tilde{F})$  is asymptotically chi-squared with  $k$  degrees of freedom and the noncentrality parameter  $\delta^*-\delta$ .
- (iii) The test statistics  $n\hat{F}$  and  $n(\hat{F}-\tilde{F})$  are asymptotically independent.

In the central case, i.e. under the null hypothesis, properties (i) and (ii) are well known (Browne, 1974; Jöreskog, 1981; Lee and Bentler, 1980). The noncentral case and property (iii) are discussed in Steiger *et al.* (1985). Also, under the null hypothesis, the following properties of correctly specified

MDF estimators hold.

(iv) The MDF estimator  $\hat{\theta}$  is asymptotically efficient within the class of all MDF estimators based on the estimator  $\hat{x}$ .

(v) The asymptotic distribution of  $n^{1/2}(\hat{\theta} - \theta_0)$  is normal with a null mean vector and covariance matrix  $(\Delta' V_0 \Delta)^{-1}$ .

Our main purpose in this paper will be to establish conditions under which properties (i)-(v) hold for possibly misspecified discrepancy functions. We extend and generalize some results of Shapiro (1986), where robustness of properties (i) and (iv) has been discussed for nonsingular  $r$ . In the analysis of covariance structures the covariance matrix  $\Sigma$  is symmetric and thus some of the elements of  $\xi = \text{vec}(\Sigma)$  are equal. This implies that the matrix  $r$  is singular with rank less than or equal to the number  $p(p+1)/2$  of nonduplicated elements of  $\Sigma$ . This motivates us to consider the case of a singular matrix  $r$ . Then we say that the discrepancy function is correctly specified if  $V_0$  is equal to a generalized inverse  $r^-$  of  $r$ , that is  $r V_0 r = r$ . It will be assumed that the mean vector  $\mu$  and the  $m \times q$  Jacobian matrix  $\Delta = (\partial/\partial \theta')g(\theta_0)$  belong to the linear space  $M(r)$  generated by the column vectors of  $r$ .

Condition (\*)

$\mu \in M(r)$  and  $M(\Delta) \subset M(r)$ .

For example, in the analysis of covariance structures condition (\*) holds if  $r$  has the maximal rank  $p(p+1)/2$ . Also,

under the assumption of multivariate normality of the data, the matrix  $F$ , denoted by  $F_N$ , becomes

$$(2.2) \quad F_N = 2M_p(\mathbf{z}_0 \otimes \mathbf{z}_0),$$

where  $\otimes$  denotes the Kronecker product of two matrices and  $M_p$  is the symmetric idempotent pattern matrix of rank  $p(p+1)/2$  (see Browne, 1974). A generalized inverse of  $F_N$  is

$$(2.3) \quad V_0 = \frac{1}{2}(\mathbf{z}_0^{-1} \otimes \mathbf{z}_0^{-1}).$$

The weight matrix  $V_0$ , given by (2.3), is associated with the standard discrepancy functions of maximum likelihood and generalized least squares (cf. Browne, 1974).

Basic asymptotics of the MDF analysis can be related to the least squares theory of linear models and described as follows. Consider the parameter set

$$\mathcal{P} = \left\{ \xi : \xi = g(\theta), \theta \in \mathbb{R}^q \right\},$$

corresponding to the model (2.1), and the min-function

$$F_{\min}(\mathbf{x}) = \min_{\xi \in \mathcal{P}} F(\mathbf{x}, \xi).$$

A second-order approximation of  $F_{\min}$  near the point  $\xi_0$  is given by

$$F_{\min}(\mathbf{x}) = \min_{\tau \in T} Q(\mathbf{x} - \xi_0; \tau) + o(\|\mathbf{x} - \xi_0\|^2),$$

where  $Q(\mathbf{y}, \tau) = (\mathbf{y} - \tau)' V_0 (\mathbf{y} - \tau)$  is the quadratic function corresponding to the second-order Taylor approximation of  $F$  and

$$T = M(\Delta) = \{ \tau : \tau = A\beta, \beta \in \mathbb{R}^q \}$$

is the linear space representing the tangent space to  $\mathcal{P}$  at  $\xi_0$ . A rigorous proof of this result can be found in Shapiro (1985b, Lemma 3.1). By the definition  $\hat{F} = F_{\min}(\hat{x})$  and hence it follows that

$$n\hat{F} = \min_{\tau \in T} Q(x^*, \tau) + o(\|x^*\|^2),$$

where  $x^* = n^{1/2}(\hat{x} - \xi_0)$ . Since  $x^*$  is asymptotically  $N(\mu, \Gamma)$  we obtain that  $n\hat{F}$  has asymptotically the same distribution as the distribution of the minimum of  $Q(y, \tau)$  subject to  $\tau \in T$ , with  $y$  being an  $N(\mu, \Gamma)$  normal variable.

Similarly, denote by  $\bar{\theta}(x)$  the (unrestricted) minimizer of  $F(x, g(\cdot))$  over  $\mathbb{R}^q$ . Clearly  $\hat{\theta} = \bar{\theta}(\hat{x})$  and  $\bar{\theta}(\xi_0) = \theta_0$ . Then

$$\bar{\theta}(x) - \theta_0 = \bar{\beta}(x - \xi_0) + o(\|x - \xi_0\|),$$

where  $\bar{\beta}(y)$  is the minimizer of  $Q(y, A\beta)$ ,  $\beta \in \mathbb{R}^q$  (Shapiro, 1985b, Lemma 3.3). It follows that

$$n^{1/2}(\hat{\theta} - \theta_0) = \bar{\beta}(x^*) + o(\|x^*\|)$$

and hence  $n^{1/2}(\hat{\theta} - \theta_0)$  is asymptotically distributed as the unrestricted minimizer of  $Q(y, A\beta)$ , with  $y \sim N(\mu, \Gamma)$ .

Throughout the paper we use the concept of orthogonal complement matrices. It is said that an  $m \times k$  matrix  $A$  is an orthogonal complement of an  $m \times s$  matrix  $B$  if  $A$  has full

column rank  $k$ ,  $k = m - \text{rank}(B)$ , and  $A'B = 0$ .

### 3. ROBUSTNESS PROPERTIES OF THE MDF TEST STATISTICS

In this section we investigate robustness properties of the unrestricted MDF test statistic  $\hat{nF}$ . We assume that the weight matrix  $V_0$  is nonsingular and denote by  $\Phi$  an orthogonal complement of the matrix  $\Delta$ . It will not be required that  $\Delta$  should have full column rank  $q$ . We rather assume that  $g(\theta)$  is regular at  $\theta_0$  in the sense that the set  $\mathcal{P}$  is a smooth manifold near  $\theta_0$  (cf. Shapiro, 1986). The restricted case can be reduced to an unrestricted one by a suitable reparameterization. Let  $Z$  be the  $q \times k$  Jacobian matrix  $(\partial/\partial\theta)c(\theta_0)$  of the constraints and  $E$  be an orthogonal complement of  $Z$ . Then all results and conditions of this section applied to  $\hat{nF}$  can be translated into the corresponding results and conditions for  $\tilde{nF}$  if one replaces the Jacobian matrix  $\Delta$  by the matrix  $\Delta E$  which corresponds to the reparametrized model.

Let the discrepancy function be given and suppose that  $r_0$  is a nonnegative definite matrix such that  $V_0 = r_0^-$ . That is the discrepancy function is correctly specified with respect to  $r_0$ . Then the following result is a generalization of proposition 3.2 in Shapiro (1986). Notice that the minimum of  $Q(y, r)$  subject to  $r \in T$  is given by the quadratic form  $y'Uy$ , where  $U$  is a nonnegative definite matrix defined alternatively by

$$(3.1) \quad U = V_0 - V_0 \Delta (\Delta' V_0 \Delta)^{-1} \Delta' V_0$$

or

$$(3.2) \quad U = \Phi (\Phi' V_0^{-1} \Phi)^{-1} \Phi'.$$

Theorem 3.1

Suppose that the discrepancy function is correctly specified with respect to a nonnegative definite matrix  $\Gamma_0$ ,  $M(\Gamma) = M(\Gamma_0)$  and condition (\*) holds. Then  $\hat{nF}$  is asymptotically chi-squared with

$$(3.3) \quad df = \text{rank}(\Gamma) - \text{rank}(\Delta)$$

degrees of freedom and the noncentrality parameter,

$$(3.4) \quad \delta = \mu' U \mu,$$

if and only if  $\Gamma$  is representable in the form

$$(3.5) \quad \Gamma = \Gamma_0 + \Delta C' + C \Delta',$$

where  $C$  is an arbitrary  $m \times q$  matrix.

Proof

First let us show that if  $V_0 = \Gamma_0^{-1}$  and condition (\*) holds, then  $\hat{nF}$  is asymptotically chi-squared. We have that  $\Gamma = BB'$ , where  $B$  is an  $m \times r$  matrix of rank  $r = \text{rank}(\Gamma)$ . Then because of the condition (\*),  $\mu$  and  $\Delta$  can be represented in the form  $\mu = B\eta$  and  $\Delta = BH$ . Consequently  $y = Bz$ , where  $z$  is  $N(\eta, I)$ , and hence

$$(3.6) \quad Q(y, \Delta\beta) = (z - H\beta)' B' V_0 B (z - H\beta).$$

It follows that  $\hat{nF}$  is asymptotically chi-squared if  $B' V_0 B = I$ . This is equivalent to  $BB' V_0 BB' = BB'$  and hence  $\Gamma V_0 \Gamma = \Gamma$ , i.e.

$V_o = r^-$ . The corresponding degrees of freedom and the noncentrality parameter are

$$df = r - \text{rank}(H) = \text{rank}(r) - \text{rank}(\Delta)$$

and

$$\eta' [B' V_o B - B' V_o B H (H' B' V_o B H)^{-1} H' B' V_o B] \eta = \mu' U \mu,$$

respectively.

Now suppose that  $V_o = r_o^-$  and show that (3.5) is necessary and sufficient for the asymptotic chi-squaredness of  $\hat{nF}$ . We have that if  $\mu \in M(r)$ , then  $y' U y$  is asymptotically chi-squared iff

$$(3.7) \quad U r U r = U r$$

(e.g. Styan, 1970; Rao and Mitra 1971). After some calculations it follows from (3.2) that (3.7) is equivalent to

$$(3.8) \quad \Phi' r \Phi (\Phi' V_o^{-1} \Phi)^{-1} \Phi' r \Phi = \Phi' r \Phi.$$

Since  $\Phi' r \Phi = \Phi' r_o \Phi$  we obtain that the sufficiency follows. Also since  $M(r) = M(r_o)$  we have that  $\text{rank}(r) = \text{rank}(r_o)$  and hence the corresponding degrees of freedom are given by (3.3).

On the other hand suppose that  $\hat{nF}$  is asymptotically chi-squared. Similarly to the first part of the proof we have that  $r_o = B B'$ ,  $\mu = B \eta$ ,  $\Delta = B H$ ,  $y = B z$  and (3.6) follows.

Moreover, since  $V_o = r_o^-$  we obtain that  $B B' V_o B B' = B B'$  and hence  $B' V_o B$  is the  $r \times r$  identity matrix  $I$ . Condition  $M(r) = M(r_o)$  implies that  $r = B \Omega B'$ , where  $\Omega$  is an  $r \times r$

positive definite matrix. Then it follows from (3.6) and proposition 3.2 of Shapiro (1986) that

$$(3.9) \quad \Omega = I + HC_1' + C_1H'.$$

Premultiplying both sides of (3.9) by  $B$  and postmultiplying by  $B'$  we obtain (3.5) with  $C = BC_1$ . ■

Note that it follows from the proof of theorem 3.1 as well as from the condition  $M(r) = M(r_0)$ , that matrix  $C$  in (3.5) can be an arbitrary  $m \times q$  matrix such that  $M(C) \subset M(r)$ . The additional condition  $M(r) = M(r_0)$  is not really restrictive. It will be satisfied if  $r$  and  $r_0$  are both of the maximal rank. The noncentrality parameter  $\delta$  given by (3.4) is independent of the matrix  $r$  and can be expressed alternatively as follows

$$\delta = \lim_{n \rightarrow \infty} \{n \min_{\theta} F(\xi_{0,n}, g(\theta))\}$$

(cf. Steiger *et al.*, 1985).

Theorem 3.1 applied to the analysis of covariance structures yields the following result.

### Corollary 3.2

Suppose that the discrepancy function is such that the associated weight matrix  $V_0$  is given by (2.3) and let the  $p^2 \times p^2$  matrix  $r$  be of the maximal rank  $p(p+1)/2$ . Then  $\hat{nF}$  is asymptotically chi-squared with

$$df = p(p+1)/2 - \text{rank}(\Delta)$$

and the noncentrality parameter  $\delta$ , if and only if  $r$  is

representable in the form

$$r = r_N + \Delta C' + C \Delta',$$

where  $r_N$  is defined by (2.2).

#### 4. ROBUSTNESS PROPERTIES OF THE DIFFERENCE TEST STATISTIC

In this section we study robustness of the difference test statistic  $n(\tilde{F} - \hat{F})$  with respect to the properties (ii) and (iii). We have that  $g(\theta)$  maps the parameter set  $\{\theta : c(\theta) = 0, \theta \in \mathbb{R}^q\}$ , corresponding to the restricted model, onto a subset  $\mathcal{P}^*$  of the set  $\mathcal{P}$ . It will be assumed that  $\mathcal{P}^*$  is a smooth manifold in a neighbourhood of the point  $\xi_0 \in \mathcal{P}^*$ . This can be ensured by essentially the same conditions as in the unrestricted situation. Then similarly to the unrestricted case,  $\mathcal{P}^*$  has at  $\xi_0$  the tangent space  $T^* = \mathcal{M}(\Delta E)$ , which is a subspace of the tangent space  $T$ , and  $n(\tilde{F} - \hat{F})$  has asymptotically the same distribution as the distribution of

$$(4.1) \quad \min_{\tau \in T^*} Q(y, \tau) - \min_{\tau \in T} Q(y, \tau)$$

with  $y \sim N(\mu, r)$ . The matrix  $E$  presents an orthogonal complement of  $Z$ . One may note that dimensions  $d = \dim(T)$  and  $d^* = \dim(T^*)$  of spaces  $T$  and  $T^*$  are given by  $d = \text{rank}(\Delta)$  and  $d^* = \text{rank}(\Delta E)$ .

Let  $A$  be an orthogonal complement of the matrix  $(\Delta E, V_0^{-1} \Phi)$ . Since  $\Phi$  is an orthogonal complement of  $\Delta$  we have that

$$\text{rank}(\Lambda) = m - \text{rank}(\Delta E) - \text{rank}(\Phi) = \text{rank}(\Lambda) - \text{rank}(\Delta E) = d - d^*.$$

Consider the matrix  $K = (\Phi, \Lambda)$ . It follows from the definitions that  $K$  is orthogonal to  $\Delta E$  and has the rank

$$\text{rank}(K) = \text{rank}(\Phi) + \text{rank}(\Lambda) = m - \text{rank}(\Delta E).$$

Consequently  $K$  represents an orthogonal complement of  $\Delta E$ . We know that the minimum of  $Q(y, \tau)$  subject to  $\tau \in T$  is given by  $y' \Phi (\Phi' V_O^{-1} \Phi)^{-1} \Phi' y$ . In a similar way the minimum of  $Q(y, \tau)$  subject to  $\tau \in T^*$  is  $y' K (K' V_O^{-1} K)^{-1} K' y$ . Then using the equality  $\Lambda' V_O^{-1} \Phi = 0$ , the difference (4.1) can be represented in the form

$$(4.2) \quad \min_{\tau \in T^*} Q(y, \tau) - \min_{\tau \in T} Q(y, \tau) = y' \Lambda (\Lambda' V_O^{-1} \Lambda)^{-1} \Lambda' y.$$

The right-hand side expression in (4.2) is a quadratic form similar to  $y' U y$ . Therefore it is possible to apply some results from the unrestricted case simply replacing  $\Phi$  and  $\Lambda$  by  $\Lambda$  and  $(\Delta E, V_O^{-1} \Phi)$ , respectively. In particular we obtain the following result (see Shapiro, 1986, proposition 3.2).

Proposition 4.1

If  $r$  is nonsingular, then  $n(\hat{F} - \tilde{F})$  is asymptotically chi-squared, with  $df = d - d^*$  and the noncentrality parameter  $\delta^* - \delta$ , if and only if

$$(4.3) \quad r = V_0^{-1} + \Delta E C_1' + C_1 E' \Delta' + V_0^{-1} \Phi C_2' + C_2 \Phi' V_0^{-1},$$

where  $C_1$  and  $C_2$  are arbitrary matrices of an appropriate order.

Note that when  $C_2 = 0$ , condition (4.3) becomes a necessary and sufficient condition for the asymptotic chi-squaredness of the restricted test statistic  $n\tilde{F}$ . Moreover, condition  $M(\Delta) \subset M(r)$  implies that  $M(\Delta E) \subset M(r)$  and hence we obtain from theorem 3.1:

Corollary 4.2

Suppose that assumptions of theorem 3.1 hold. Then  $n\hat{F}$  and  $n(\hat{F} - \tilde{F})$  are both asymptotically chi-squared if and only if  $r$  is representable in the form

$$(4.4) \quad r = r_0 + \Delta E C' + C E' \Delta',$$

where  $C$  is an arbitrary matrix of an appropriate order.

Now let us consider property (iii). We have that  $n\hat{F}$  and  $n(\hat{F} - \tilde{F})$  are asymptotically independent iff the quadratic forms  $y'Uy$  and  $y' \Lambda (\Lambda' V_0^{-1} \Lambda)^{-1} \Lambda' y$  are independently distributed. This leads to the following necessary and sufficient condition (e.g. Styan, 1970; Rao and Mitra, 1971, p. 178):

$$\Lambda (\Lambda' V_0^{-1} \Lambda)^{-1} \Lambda' r \Phi (\Phi' V_0^{-1} \Phi)^{-1} \Phi' = 0,$$

which is equivalent to

$$(4.5) \quad \Lambda' r \Phi = 0.$$

We obtain the following result.

Theorem 4.3

The statistics  $n\hat{F}$  and  $n(\hat{F}-\tilde{F})$  are asymptotically independent if and only if condition (4.5) holds.

Condition (4.5) is equivalent to  $M(\Gamma A) \subset M(\Delta)$ , which in turn means that

$$(4.6) \quad \Gamma A = \Delta X$$

for some matrix  $X$  of an appropriate order. By the definition of  $A$ ,  $A'V_O^{-1}\Phi = 0$  which implies that there exists a matrix  $P$  such that  $A = V_O\Delta P$ . Substituting that into (4.6) we obtain

$$(4.7) \quad \Gamma V_O\Delta P = \Delta X.$$

If  $V_O = \Gamma^{-1}$  and  $M(\Delta) \subset M(\Gamma)$ , then  $\Gamma V_O\Delta = \Delta$  (e.g. Rao and Mitra, 1971, p. 21) and hence (4.7) holds with  $X = P$ . Therefore if the discrepancy function is correctly specified and  $M(\Delta) \subset M(\Gamma)$ , then  $n\hat{F}$  and  $n(\hat{F}-\tilde{F})$  are asymptotically independent. Moreover, if  $A'\Gamma_O\Phi = 0$  for some  $\Gamma_O$ , then (4.5) holds for

$$(4.8) \quad \Gamma = \alpha\Gamma_O + \Delta E C' + C E' \Delta' + V_O^{-1} \Phi Q \Phi' V_O^{-1},$$

where  $C$  and  $Q$  are arbitrary matrices with  $Q$  being symmetric and  $\alpha$  is an arbitrary nonnegative number.

Corollary 4.4

Suppose that the discrepancy function is correctly specified with respect to a nonnegative definite matrix  $r_o$  and  $M(\Delta) \subset M(r_o)$ . Then  $\hat{nF}$  and  $\hat{n}(\tilde{F}-F)$  are asymptotically independent if  $r$  is representable in the form (4.8).

The results of corollaries 4.2 and 4.4 can be applied straightforwardly to the analysis of covariance structures if the discrepancy function satisfies condition (2.3),  $r_o = r_N$  and  $r$  has the maximal rank  $p(p+1)/2$ .

## 5. ROBUSTNESS PROPERTIES OF THE MDF ESTIMATORS

In this section we study asymptotic efficiency of the MDF estimator  $\hat{\theta}$ . For the sake of simplicity we suppose, in addition to the assumptions of section 2, that the matrix  $\Delta$  is of full rank  $q$  and the model holds exactly. This implies that  $\hat{\theta}$  is uniquely defined in a neighbourhood of  $\theta_o$  for all  $\hat{x}$  sufficiently close to  $t_o = g(\theta_o)$ . Moreover,  $\mu = 0$  and hence condition (\*) becomes  $M(\Delta) \subset M(r)$ .

As we have mentioned in section 2,  $n^{1/2}(\hat{\theta} - \theta_o)$  is asymptotically distributed as the unrestricted minimizer  $\bar{\beta}$  of  $Q(y, \Delta\beta)$ , with  $y \sim N(0, r)$ . It is well known that  $\bar{\beta} = (\Delta'V_o\Delta)^{-1}\Delta'V_o y$  and hence  $n^{1/2}(\hat{\theta} - \theta_o)$  is asymptotically normal with a null mean vector and the covariance matrix

$$(5.1) \quad \Sigma = (\Delta'V_o\Delta)^{-1}\Delta'V_orV_o\Delta(\Delta'V_o\Delta)^{-1}$$

(cf. Browne, 1974, Proposition 2; Shapiro, 1983, p. 65). If  $M(\Delta) \subset M(r)$ , then  $\Delta'r^{-1}\Delta$  is independent of a particular choice

of  $\Gamma^{-1}$  (e.g. Rao and Mitra, 1971, Lemma 2.2.4) and is a positive definite matrix. By the Gauss-Markov theorem

$$(5.2) \quad \Pi \geq (\Delta' \Gamma^{-1} \Delta)^{-1}$$

in the Loewner sense of inequality, i.e.  $\Pi - (\Delta' \Gamma^{-1} \Delta)^{-1}$  is a nonnegative definite matrix (cf. Browne, 1974, Proposition 3; Shapiro, 1986, Proposition 5.1). The estimator  $\hat{\theta}$  is said to be asymptotically efficient (within the class of all MDF estimators based on the estimator  $\hat{x}$ ) if  $\Pi = (\Delta' \Gamma^{-1} \Delta)^{-1}$ . It follows from the Gauss-Markov theorem and the discussion above that asymptotically efficient MDF estimators are asymptotically equivalent to each other. That is if  $\hat{\theta}$  and  $\hat{\theta}^*$  are two asymptotically efficient MDF estimators associated with the same model and the same estimator  $\hat{x}$ , then

$$\text{plim}_{n \rightarrow \infty} n^{1/2} (\hat{\theta} - \hat{\theta}^*) = 0.$$

Conditions under which the covariance matrix  $\Pi$  attains its lower bound given by the right-hand side expression of (5.2), have been discussed extensively (see Rao, 1967; Watson, 1967; Zyskind, 1967; Kruskal, 1968; and references therein). Applied to our situation this theory yields the following result.

Theorem 5.1

Suppose that  $M(\Delta) \subset M(\Gamma)$ . Then any one of the following equivalent conditions is necessary and sufficient for asymptotic

efficiency of  $\hat{\theta}$ :

- (a)  $\Phi'RV_0\Delta = 0$
- (b)  $M(RV_0\Delta) \subset M(\Delta)$
- (c)  $RV_0\Delta = \Delta X$  for some matrix  $X$ .

Proof

In the case of  $R$  the  $m \times m$  identity matrix conditions (a)-(c) are well known (Rao and Mitra, 1971, Theorem 8.2.1(i); Zyskind, 1967, Theorem 2(6)). In the general case, as in the proof of theorem 3.1, we can write  $R = BB'$ ,  $\Delta = BH$  and  $y = Bz$  for some  $m \times r$  matrix  $B$  of rank  $r = \text{rank}(R)$ . Then  $Q(y, \Delta\theta)$  is representable in the form (3.6) with  $z \sim N(0, I)$ . Now applying condition (a), for the identity matrix  $I$ , we obtain that  $\hat{\theta}$  is asymptotically efficient iff

$$(5.3) \quad F'B'V_0BH = 0$$

where  $F$  is an orthogonal complement of  $H$ . The orthogonal complement  $F$  can be taken in the form  $F = B'\Phi$  and hence (5.3) is equivalent to the condition (a). ■

It should be mentioned that condition (b) and  $M(\Delta) \subset M(R)$  imply that actually  $M(RV_0\Delta) = M(\Delta)$ .

Theorem 5.1 implies the following result.

Corollary 5.2

Suppose that the discrepancy function is correctly specified with respect to a nonnegative definite matrix  $R_0$ ,  $M(\Delta) \subset M(R_0)$  and  $M(R) = M(R_0)$ . Then  $\hat{\theta}$  is asymptotically efficient if and only if  $R$  is representable in the form

$$(5.4) \quad \Gamma = \alpha \Gamma_0 + \Delta \Delta' + \Gamma_0 Q Q' \Gamma_0,$$

where  $D$  and  $Q$  are arbitrary symmetric matrices and  $\alpha$  is an arbitrary nonnegative number.

### Proof

As in the second part of the proof of theorem 3.1 we have that  $\Gamma_0 = BB'$ ,  $\Gamma = B \Omega B'$ ,  $\Delta = BH$  and  $B'V_0B = I$ . Then condition (a) of theorem 5.1 reduces to  $F' \Omega H = 0$  with  $F = B'Q$ . This is equivalent to

$$(5.5) \quad \Omega = \alpha I + HDH' + FQF'.$$

(See Rao and Mitra, 1971, Theorem 8.2.1(iii)). Premultiplying both sides of (5.5) by  $B$  and postmultiplying by  $B'$  we obtain (5.4). ■

Theorem 3.1 and corollary 5.2 imply the following result.

### Corollary 5.3

Suppose that assumptions of Corollary 5.2 hold. Then  $n\hat{F}$  is asymptotically chi-squared and  $\hat{\theta}$  is simultaneously asymptotically efficient if and only if  $\Gamma$  is representable in the form

$$(5.6) \quad \Gamma = \Gamma_0 + \Delta \Delta',$$

where  $D$  is an arbitrary symmetric matrix.

In some situations the discrepancy function under consideration is known to be correctly specified with respect to a nonnegative definite matrix  $\Gamma_0$ . Then the associated asymptotic covariance matrix of  $n^{1/2}(\hat{\theta} - \theta_0)$ , for  $\Gamma = \Gamma_0$ , is given by

$\Sigma_0 = (\Delta' V_0 \Delta)^{-1}$ , i.e. property (v). On the other hand we have seen that if  $\hat{\theta}$  is asymptotically efficient, then the asymptotic covariance matrix  $\Sigma$  is  $(\Delta' R^{-1} \Delta)^{-1}$ . In the case where  $R$  is expressible in the form (5.6) there is a simple formula connecting  $\Sigma$  and  $\Sigma_0$ . First suppose that  $R, R_0$  and  $D$  are nonsingular. Then by a known matrix identity (e.g. Rao, 1973, p. 33) it follows from (5.6) that

$$(5.7) \quad R^{-1} = R_0^{-1} - R_0^{-1} \Delta (\Delta' R_0^{-1} \Delta + D^{-1})^{-1} \Delta' R_0^{-1}.$$

Applying the same matrix identity to the inverse of the matrix  $\Delta' R^{-1} \Delta$  resulting from (5.7) we obtain after some calculations that

$$(\Delta' R^{-1} \Delta)^{-1} = (\Delta' R_0^{-1} \Delta)^{-1} + D.$$

By the argument of continuity the same result holds for singular  $D$ ,  $R$  and  $R_0$  (with  $R^{-1}, R_0^{-1}$  replaced by  $R^-, R_0^-$ , respectively) if  $M(\Delta) \subset M(R_0)$ ,  $M(R) = M(R_0)$ , and hence  $\Delta' R^{-1} \Delta$  and  $\Delta' R_0^{-1} \Delta$  are independent of a particular choice of the generalized inverse. We obtain

$$(5.8) \quad (\Delta' R^{-1} \Delta)^{-1} = (\Delta' R_0^{-1} \Delta)^{-1} + D$$

and hence

Corollary 5.4

Suppose that the discrepancy function is correctly specified with respect to  $\Gamma_0$ ,  $M(A) \subset M(\Gamma_0)$ ,  $M(\Gamma) = M(\Gamma_0)$ , and  $\Gamma$  is representable in the form (5.6). Then  $\hat{\theta}$  is asymptotically efficient and

$$(5.9) \quad \Pi = \Pi_0 + D.$$

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